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Scalable Innovation Process Development in Grocery Retail: A Case Study

A Scientific Study of Grocery Retail Innovation

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Utveckling av en skalbar innovationsprocess för dagligvaruhandeln: En fallstudie

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Sammanfattning

Syftet med studien är att utveckla en skalbar innovationsprocess för ett företag i den svenska dagligvaruhandeln. I detaljhandeln verkar företag med små vinstmarginaler i en konkurrensutsatt och reglerad marknad. Detta ökar behovet av etablerade innovationsprocesser för att säkerställa långsiktigt affärsvärde. Studien syftar till att ge insikter om utmaningar förknippade med att utveckla och implementera innovationsprocesser i ett dagligvaruföretag. En iterativ litteraturstudie genomförs, samt en semistrukturerad intervjustudie, som tematiskt analyseras för att identifiera nyckelutmaningar hos fallföretaget.

Resultaten visar åtta nyckelutmaningar relaterade till innovationsarbete inom dagligvaruhandeln, inklusive ad-hoc-initiativ, organisatoriska silos samt brist på samarbete och processer för att spara idéer. En skalbar innovationsprocess i sex steg har utvecklats för att möta dessa utmaningar, med stöd av en innovationsplattform. För att möta de viktigaste utmaningarna krävs att innovationsprocessen är tillräckligt flexibel och robust för implementering i full skala hos företagets affärsenheter för att strukturera hur innovationer utvecklas och implementeras. Studiens bidrag till akademin och det praktiska arbetet inom dagligvaruhandeln omfattar branschinsikter och diskussioner om utveckling av skalbara innovationsprocesser för dagligvaruhandeln.

Nyckelord: skalbar innovationsprocess, dagligvaruhandel, empirisk fallstudie



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Abstract

The purpose of the study is to develop a scalable innovation process for a case company in the Swedish grocery retail market. In the larger retail market, companies operate with small profit margins in a competitive and regulated market. This increases the need for established innovation processes to capture value long-term. The study offers insights on challenges associated with developing and utilizing innovation processes in a grocery retail company. An iterative literature study is conducted, alongside a semi-structured interview study, which is thematically analyzed to identify key challenges in the case company.

The results showcase eight key challenges related to innovation work in retail, including ad-hoc initiatives, organizational silos as well as lack of collaboration and processes for storing ideas. A six step scalable innovation process is developed to address these challenges, supported by an Innovation Platform. To address the key challenges, the innovation process is required to be flexible and robust enough for implementation at full-scale across the case company's business units to structure the way innovations are developed and implemented. The study's contribution to academia and the practical work in grocery retail includes industry insights and discussions on the development of scalable innovation processes for grocery retail.

Keywords: scalable innovation process, grocery retail, empirical case study

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1. Introduction

1.1 Background

The Swedish retail industry is one of the most crucial for the national economy, accounting for 11 percent of BNP and 14 percent of the tax revenue in 2021 (Svensk Handel, 2023). Despite this, the retail industry operates with small profit margins, selling commodity fast-moving consumer goods (FMCG) in a highly competitive market. The logistics value chain is complex as operations span from production of private label products, supplier negotiations, distribution and wholesalers to consumer interactions in store and after product usage (McKinsey & Company, 2024).

To both ensure a competitive position in the future, and to comply with regulations from the European Commission and national laws, retailers are setting ambitious growth and sustainability goals (European Commission, 2019). The retail industry is therefore highly dependent on scalable innovation processes to capture and develop new ways of working long-term. In contrast to ad-hoc innovation work, structured and scalable innovation processes that can be implemented across business units can increase both the effectiveness and the profitability for retailers (Jonsdottir, 2024). As suggested by Trott (2020), local and ad-hoc innovations have a limited organizational impact and there is an alternative cost of not implementing scalable ideas across different business units and levels of a company. Additionally, the lack of processes for implementing innovations in a structured manner is associated with the risk of losing organizational knowledge (Riege, 2005). This also causes missed opportunities for synergy effects between business units as suggested by Crossan & Apaydin (2010).

Grocery retail is an important sub-sector of the retail industry. In the nature of distributing and selling FMCG, the high turnover rates requires the grocery retail companies to quickly adapt to changes in both supply chain operations and consumer behavior by constantly streamlining workflows and seeking out new best ways of working. To balance cost effectiveness driven by the low margin FMCG as well as high customer demands, the importance of identifying and

implementing innovations are crucial to stay competitive (Fernie & Sparks, 2018).

In the Swedish grocery retail market, there is an oligopoly with few large actors having a majority of the market share. The inconsistent innovation process nature in the market is driven by the competitive landscape, where companies set five year strategies for operational work, and have capacity to implement few short and ad-hoc innovations as response to emerging customer needs and/or competitive initiatives in the market (OECD, 2016).

Innovation processes, as defined by Tidd & Bessant (2020) are structured frameworks for:

- Developing and implementing new ideas for market diffusion e.g. developing a service or product.
- Preserving value for a company.

These processes typically start with the need for a complex change, and result in an implementation of a solution for the initial problem, and therefore differ from idea generation where the structured implementation is not methodologically included (Smith, 2003). Additionally, a report from Binns et al. in 2021 states that in Europe, food production is responsible for 20-30% of greenhouse gasses. As suggested in a study by Ortiz-de-Mandojana & Bansal in 2015, companies who work with sustainable innovation in scalable ways are more likely to achieve long-term organizational resilience and improved financial volatility, which are key to remaining competitive in a low margin industry.

To ensure financial competitiveness and comply with sustainability regulations, retailers are dependent on innovation work to stimulate and streamline new ways of optimizing their value chains, products, operations and business models. While profitability is a key driver for most retailers, sustainability work is becoming more central in business strategies for long-term growth. However, despite the ongoing adoption of innovation and sustainability across the retail industry, there are opportunities to further understand and explore the development of structured innovation processes. Saxena et al. (2024) argues that structured processes are essential for fostering sustainable innovation. A lack of formal systems can therefore affect the adaptability to sustainable business models and make it difficult to maintain and replicate sustainable innovation over time. Today, the

industry specific academic literature on innovation processes in the Swedish grocery retail market is limited, as many studies have a broader geographic and/or retail industry focus.

1.2 Purpose

The purpose of the study is to develop a scalable innovation process to support the case company's strategy and competitiveness in the grocery retail sector with tight margins and constant change. The innovation processes should take into account the identified key challenges for innovation process work in the grocery retail case company, limiting ad-hoc silo work and ensuring process repeatability. Additionally, the study will research key steps in established innovation processes, and address these steps for the development of a tailored innovation process to meet the needs of the case company. The study aims to combine insights from the company case study with scientific learnings on innovation processes.

Research questions:

- RQ1. Which key challenges and features characterize the innovation process work in a grocery retail company?
- RQ2. What does an innovation process look like to be scalable and applicable across different business units in a grocery retail company?

1.3 Delimitations

The study investigates one large company's practical innovation work in the Swedish grocery retail market, to develop a scalable innovation process tailored to their needs. The study is centered around the current work and development of a new innovation process, and does not consider the implementation phase. Moreover, the study primarily considers the grocery retail industry, however the literature study includes additional review of other retail industry segments. In the data collection through an interview study, the interview objects work with development in different business units and capacities within the case company and its corporate group organization. Geographically, the case company is operational in the Swedish market.

2. Literature study

2.1 Retail and innovation

Historically, the retail sector has been essential in assuring access to goods and services for customers. In today's competitive market, retailers must sustain growth and address challenges by enhancing flexibility, efficiency and innovation. Additionally, it is crucial for retailers to recognize their role in driving industry innovation, and to identify and develop scalable innovation processes internally in the ongoing process. According to Howells et al. (2004), retailers are focusing more on service innovations to stay competitive in the industry. This understates the importance for retailers to iterate business models and processes, making sure to innovate beyond product offerings in order to align with customer needs. However, according to a report by OECD (2005) less than 40% of companies in the retail industry consider themselves to be innovative. Together with the communication and transport industries, retail is considered one of the least innovative industries based on self evaluation from companies.

Two key drivers of success for companies in the retail sector are creating customer value, and how well the product offering meets customer needs. According to Holbrook (1994), customer value is a subjective experience occurring when a customer interacts with a product offering or service. It includes various metrics to determine value, including financial, social and emotional, driving the overall interaction experience. In addition, customer needs are commonly segmented into social, economic and functional needs fulfilled when the customer interacts with the product according to a study by Spivak.

Therefore, for a grocery retail company to be successful in terms of adhering to customer needs and creating value, it is crucial to drive internal innovation to create customer-centric offerings and differentiate from competitors. Compared to industries like healthcare, energy and engineering services, the retail sector is less advanced in innovation with fewer radical developments and less technical height in product offering advancements. One contributing factor is that innovation traditionally is measured using standard

metrics, such as number of patents, product innovation and revenue share (Sundström & Radon, 2014), where innovations in retail are less visible and harder to quantify compared to industries with more conventional R&D processes. According to Hristov & Reynolds (2009), one of the reasons why fewer innovations are reported in retail is their approach to innovation processes. Furthermore, fewer technical advances can influence how merchants evaluate their innovation capabilities, thereby underestimating their level of innovation.

However, according to Howell & Tether (2004), rather than relying on technological development, retail innovation often consists of incremental changes in processes, with significant transformation occuring every few decades. While advancements such as changes in store layout, launching of new private labels and loyalty programs, more substantial innovations are rare and tend to occur only every few decades. While the retail sector changes over time, the pace of more radical innovation is slower than other industries (Salsberg et al., 2015).

A report from the European Union (2015), states the retail sector is wrongly perceived as poorly innovative despite it being recognised as highly competitive. To provide customers with the right products and services, retailers engage in daily competition. As a result, they have to constantly innovate to keep pace with rapidly changing demands and an evolving market environment. Compared to other sectors, retailers are more common to engage in open, collaborative innovation. When developing innovative solutions, it is standard practice for retailers to incorporate both internal and external ideas that often incorporate collaboration with partners from the supply chain. By doing so, retailers can co-create value with suppliers. Furthermore, the retail sector engages in both technological and non-technological innovation. Overall, there is a big focus on developing new SaaS solutions, such as loyalty marketing systems, payment methods and mobile platforms, as well as non-technological innovations e.g., new business models and marketing approaches (European Union, 2015).

As mentioned above, modern retail innovation is not only driven by efficiency improvements, but more importantly by enhancing the overall customer experience, including value, honesty, convenience, transparency and omni-channel services. Here it becomes crucial for retail leadership to identify new ways of creating customer value, and at the same time generate revenue from any surplus value. As a

result, innovation in retail has to be a blend of scientific retail innovation and creative processes. Fundamentally, retail innovation must significantly improve the quality of the entire shopping experience before, during and after purchase. For this to be possible, it requires a deep understanding of the buying process, identifying potential barriers and offering solutions that provide value to the customer. The creative process essential for innovation requires empathy for customers' lifestyles, decisions and expectations. However, understanding customer decision-making and drawing meaningful conclusions from them, necessitates cooperation with scientific researchers (Sundström & Radon, 2014).

Innovation in the retail sector can be either radical, with noticeable changes that are easy to recognize by consumers, or incremental, adding up continuous improvement to result in improved products, processes and organizational methods. The retail sector innovation accelerates more rapidly than consumers' needs evolve, and therefore innovation sometimes becomes indistinguishable in consumers' eyes. Ultimately, recognizing disruptive innovation can be difficult, making the incremental innovation even more difficult to acknowledge. Incremental innovations often go unnoticed by customers, yet they lead to significant benefits and savings (European Union, 2015).

2.2 Innovation process frameworks

Several well-established innovation processes for structuring the innovation work in different professional environments exist. Three of the most established processes in innovation process literature include:

The Stage Gate Process developed by Cooper in 1990 (Figure 1). It contains five strategic stages for project related activities, and gates for decision making before moving on to the next stage. This methodology is designed to financially de-risk implementations and enhance productivity in product development environments. Several methodologies inspired from Coopers' process, including the Nextgen Stage Gate which takes market fluctuations into account (DelVecchio et al., 2013).

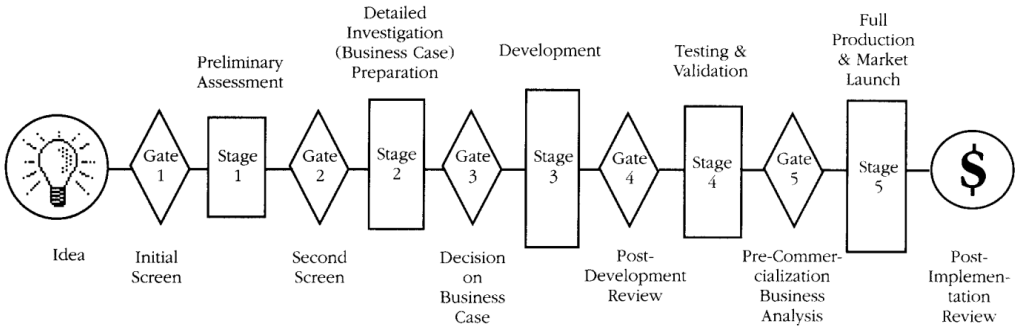


Figure 1. The Stage Gate Process. Reprinted from Stage-gate systems: A new tool for managing new products (p. 45) by Cooper, 1990. Business Horizons Vol 33.

Lean Startup developed by Ries in 2011 is centered around hypothesis based development to reduce waste and maximize value (Figure 2). This allows for rapid iterations, minimum viable product introduction and early feedback and therefore de-risk the development process (Gamon-Sanz et al., 2024).

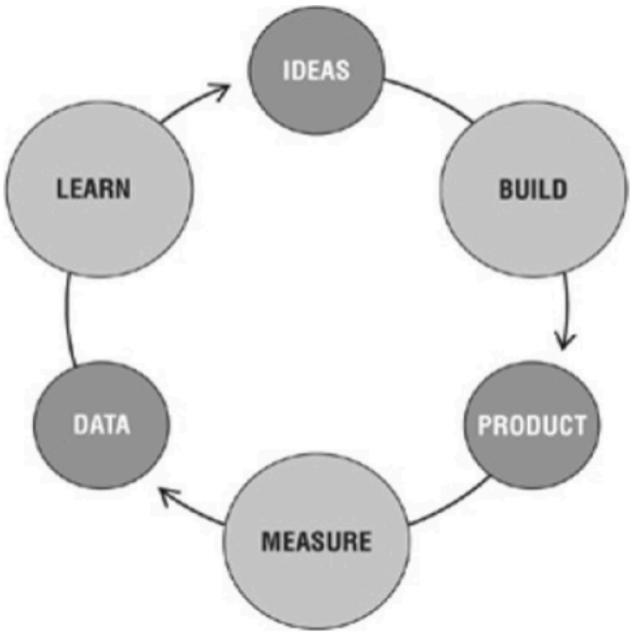


Figure 2. Lean Startup. Reprinted from The Lean Startup: How Today's Entrepreneurs Use Continuous Innovation to Create Radically Successful Businesses (p. 75) by Ries, 2011.

Design Thinking methodologies are user-centric ways of driving innovation processes in organizations, with a strong emphasis on empathy and feedback (Liedtka, 2018).

1. Customer Discovery	
1.1. Immersion	Immersion (in the customer experience) produces data, transformed into insights, which help teams agree on design criteria to brainstorm solutions.
1.2. Sense-making	The process of creatively analysing the information gathered from immersion e.g. Gallery Walk model.
1.3. Alignment	Series of seminar discussions that focus on possibilities, helping diverse teams have more-collaborative discussions about the design criteria of ideal design.
2. Idea Generation	
2.1. Emergence	Establishing dialogue about potential solutions, planning who will participate, delegating responsibility, and the structure of the conversation.
2.2. Articulation	Assumptions about what's critical to success of those solutions are examined.
3. Testing Experience	
3.1. Pre-experience	Solutions tested w/ rough prototypes that help teams further develop designs.
3.2. Learning in Action	Real-world experiments are an essential way to assess new ideas and identify the changes needed to make them workable.

Figure 3. Why Design Thinking Works (p. 78) by Liedtka, 2018. Harvard Business Review Vol 96.

The Process of Agile Innovation as described by Lokuge (2015) calls for an organizational innovation platform where all sources and information is consolidated (Figure 4). Additionally, the focus is centered around engagement from several internal and external stakeholders to ensure that flexible control structures can be leveraged to measure progress (Krstic et al., 2018).

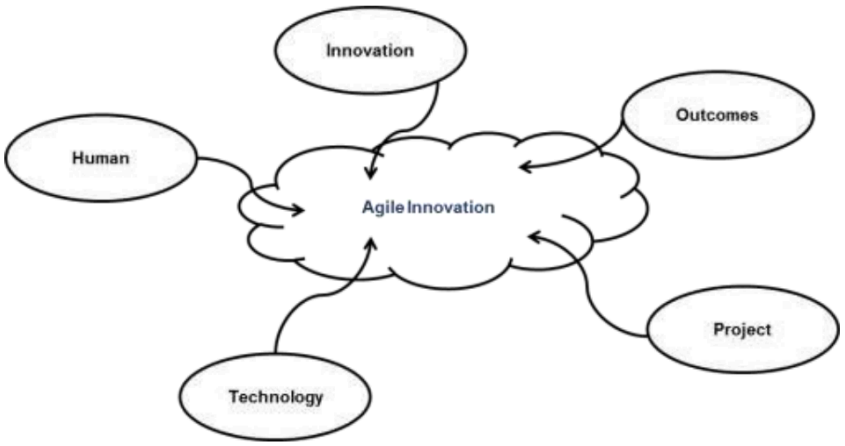


Figure 4. Agile Innovation (p. 164) by Lokuge, 2015. Agile innovation: Innovating with enterprise systems. PhD thesis, Queensland University of Technology.

2.3 Historical overview: ad-hoc to systematic innovation process in retail

The historical innovation process work in retail has shaped the way the industry meets customer demands and adjusts to trends in society today. In most instances, the innovation processes to implement new products, services or ideas within grocery retail has been conducted in an ad-hoc and case-driven manner, rather than by utilizing structured innovation processes to develop and implement change. This ad-hoc way of developing and implementing innovations is characterized by replications of current solutions performed by competitors, rather than driving systematic innovation based on company needs via scalable processes (Sheu & Lee, 2010). As a cause, innovation work is subject to silo effects where teams reinvent the wheel independently, rather than leveraging established company processes to execute on their work on innovation. Therefore, Sheu and Lee are advocating for the implementation of Systematic Innovation Processes (SIP), which is defined as a set of stages and phases where systematic innovation tools are utilized. The need for more systematic innovation processes in grocery retail is further strengthened by Lopes (2022), who claims that the utilization of innovation processes in large organizations enables innovation technologies to be developed. Additionally, Lopes argues that the result can be more sustainable manufacturing and distribution conditions, adhering to a higher level of ESG compliance in production whilst streamlining internal process work. To summarize, there is a clear split between technical product innovation and the process improvements to implement innovations, driving change across the value chain.

In retail, a lot of ad-hoc innovations have been implemented throughout the 20th century (Kerssens, 2017). Retail innovations have typically been incremental rather than radical, causing retail players to miss out on opportunities and hence causing great alternative costs according to Evans (2011). By understanding the historical innovations in the industry, the successes and challenges can be better understood. However, as the society is evolving and becoming more digitized, new means of innovations will be utilized to drive enduring change as the business landscape changes.

Technologically, barcodes were a technological innovation introduced to the retail industry in the 70s, enabling effectivization of the payment processes in stores as well as warehouse operations. Moreover, electronic cashier systems and contactless payment solutions like RFID further enabled this throughout the 80s and 90s. Schnedlitz et al (2010) claims that a wide range of payment solutions to facilitate the checkout process for customers increases the shopping experience and enables more loyalty services.

In the early 21st century, process innovations of the customer behavior in store like self-checkouts, digital mobile applications and loyalty systems were introduced, enabling a more personalized experience and more warehouse predictions (Rivera et al., 2021). For sustainability work, these innovations have all contributed by enabling increased waste management insights and operational efficiency.

2.4 Innovation and innovation processes in retail today

Two themes regarding the development and implementation of innovations are detected in research by Pantano & Giuseppe (2012) and Davies (2006), where their descriptions reflect the retail industry today. Technology is being integrated into the customer experience to a large extent, and virtual reality solutions are becoming more frequent in the industry. Moreover, as retailers have presence both in physical stores and online, the innovation processes and the product development are becoming more conjoined.

Another trend in modern retail is a clear shift from mostly hardware and firmware innovation to the development of customer-centered interactions. According to Hermes & Riedl (2021), this aims to retain engagement and strengthen the relationship to the customer, rather than only utilizing emerging technologies to effectivize and optimize the operations of store logistics. Hence, innovation processes to create attractive environments and loyalty strategies are becoming more of a must-have than a nice-have (Rodriguez et al., 2016).

Whilst the common themes of technology integration is to enhance customer experiences and the customer-centric innovation processes to ensure enduring loyalty, there are some contrary

conclusions to the researchers. Davies (2006) argues that the operational effectiveness for retailers by standardizing processes and therefore creating opportunity for innovations like cashier advancements, has been the largest change in modern day retail. However Pantano & Giuseppe (2012) suggests that the largest focus on retail has been the emerging technologies being implemented to create more interactive customer journeys.

Despite the differing perspectives of drivers of retail innovation, one central aspect remains consistent: the customer centered approach. Aligning the innovation processes and efforts with customer preferences and expectations is something retailers are becoming increasingly more aware of. A study conducted by Albors-Garrigos (2020) shows how innovation has leaned towards customer-centric strategies, and a leader of co-innovation initiatives is the major Spanish grocery retailer, Mercadona. They bring customers directly into the innovation process using established co-innovation labs, allowing customers to contribute with ideas after using the product in home-environments. This method of collaborating with customers in the innovation process has fostered more feedback and a more customized product development approach.

Besides collaborating with the consumers, the externally collaboration with suppliers plays a key role in the innovation process. By the integration of suppliers, new products and services can be developed that meet evolving needs and as expertise and resources from the suppliers are used, retailers can commercialize products more effectively. By integrating suppliers into the innovation process, Mercadona could generate 30% more new ideas in their co-innovation labs (Albors-Garrigos, 2020).

For innovation processes in modern retail, Zolnowski & Christiansen (2016) argues that rather than having scalable innovation processes, retail is dependent on ad-hoc, non-systematic workflows to retain innovations. One driver for this is the focus on innovation strategies rather than innovation processes, causing a larger focus on quick strategic iterations rather than long term innovation process development, causing fragmentation and less systematic approaches.

In grocery retail, innovation work is becoming increasingly important to manage through technical advancements, digitalization trends and

increasing customer demands. Through recent years, Industry 4.0 technologies from the manufacturing industry have revolutionized operational efficiency. Internet of Things, big data analysis and digital twin systems optimizes business models and enable players to adapt faster to meet market demands. Hence, both operational efficiency and customer-centric strategies with personalized relationships are becoming more frequent in the industry (Barile et al., 2018).

Innovation processes in grocery retail are needed across the service and product life cycles. According to Rivera et al., grocery retailers can leverage IoT technologies to better understand customer interactions with the product portfolio, and hence optimize the warehousing process with RFID technologies (2021).

2.5 Innovation barriers in retail

This section presents the barriers that, according to previous research in the area, are considered the most frequently mentioned and most prominent as limiting innovation in the retail sector.

2.5.1 Market dynamics

In the retail industry, according to Hristov and Reynolds, (2015), innovation is more influenced and focused on the continuous adaptation to market changes and customer needs. Therefore, sufficient information of market trends and changes is crucial for innovation, and a lack of this information can be a significant barrier. The retail industry has less emphasis on technology change and is more concerned with responding to transforming market demands.

2.5.2 Finances

Retail margins are often low and the sector is administratively complex. As companies have to primarily invest in employees, physical space, administrative systems and project processes, it can be challenging to justify financing for radical innovation projects, or even incremental innovations (Reynolds and Sundström, 2013). In addition, Bravi et al. (2021) notes that this is a common challenge

large companies in the food sector face. Barilla for example, has challenges in managing innovation projects due to the high costs and long development times connected to traditional business models. Bravi et al. (2021) further elaborates on how Barilla, through consumer testing and scale, has created a strategy that relies on building a well-known and trusted brand. However, this approach is not working well for radical innovation or flexible adaptability in a fast-moving market. Therefore, the large upfront costs associated with implementing innovation are often a financial barrier despite the long term benefits and long-term cost saving opportunities associated.

2.5.3 Management skills and expertise

One innovation barrier discovered is the lack of customer-centric attitude within retail management teams and board members. Research by Reynolds & Sundström (2013) highlights that retail leaders often lack the ability to view the business, strategies and needs from the customer's perspective. Therefore, building a strong customer-centered culture has been identified as a key driver for innovation in the retail industry.

Furthermore, according to Larsen and Lewis (2007), the lack of management expertise is a reason for business failure and therefore a barrier to innovation. A management team that lacks qualities such as determination, skill, good judgment and experience can not lead projects to success at the same level as teams possessing these qualities. Moreover, as a leader, it is important to be able to undertake competitor analysis and collect sufficient data to monitor development performance.

Successful innovation projects often hinge on collaboration between retail business, companies from other sectors, and scientific research. However, there is significant potential for improving cooperation in training and education related to retail innovation, where barriers exist today. According to the strategic framework by European cooperation, it is crucial to enhance creativity and innovation at all levels of leadership training to provide retailers with support to overcome these barriers (Reynolds & Sundström, 2013).

2.5.4 Internal process rigidity

In addition to management skills gaps, large companies like Barilla can struggle to implement innovation processes. One reason behind this is due that they are overly focused on internal processes and being stuck in established methods, which contrasts with the fast-paced, agile approaches. Bravi et al. (2021) argues that large companies in these situations, either fail to find disruptive innovation or overextend responsibilities and resources, not addressing the goals of the innovation process. Therefore it is important for retail management to consider both the adaptability and the goals of innovation to ensure successful implementation.

A business model is a well- specified system that serves as a company's organizing logic for value creation and value appropriation. For the retail sector, the business model can be described as how the retailer creates value for the company and its customers. For the business model innovation to be successful, the focus on the customer and the ability to compete for customers are of importance. A hinder to these focuses would therefore in principle be a barrier to innovation (Reynolds & Sundström, 2013).

2.5.4 Lack of formal structures

To manage the innovation process effectively, retailers often lack the formal structures and systems needed for this. Instead, innovation tends to be more ad-hoc, which can hinder systematic progress. The approach to innovation in the retail sector is more experimental and less organized (Hirstov, 2007), and the formalization of innovation processes, inputs and outputs is limited, with more focus on non-formal innovation processes (Reynolds & Cuthbertson (2014), Olsson et al. (2019).

According to a study by Olsson et al. (2019), results show that innovation is not enabled to the extent that is needed among Swedish retail organizations. Reasons for this is the lack of support that comes in the form of formal structures, processes and resources allocated to innovation initiatives. Furthermore, the absence of formalized frameworks to guide and manage innovation effectively affects organizations, leading to employees not getting clarity nor control on how to promote innovation work, and get the feeling that

too little is said about innovation. The study also mentions that due to poor formal innovation processes and structures not being in place, employees perceive taking initiatives as less welcomed and needed (Olsson et al., 2019).

3. Method

3.1 Research design

The intention of this study is to build a perspective on how the grocery retail case company can develop a scalable innovation process tailored to addressing the identified key challenges for innovation work. To ensure that the innovation process is well-developed and adapted to fit the needs of the case company, the study is based on a comprehensive methodology including both an iterative literature review and an empirical case study (Figure 5).

The methodology allows for a flexible and adaptable research process where new information from the data collection work can be continuously integrated and analyzed. The study is based on a combination of case study and theoretical framework, which makes it possible to explore the details and complexity of the problem space to build perspective on pain points and the current situation for a scalable innovation process to be developed. As emphasized by Bryman and Bell (2017), this methodology enables an evaluation between theoretical concepts and real situations, enabling a deeper understanding of the processes and behaviors involved in the company's innovation work.

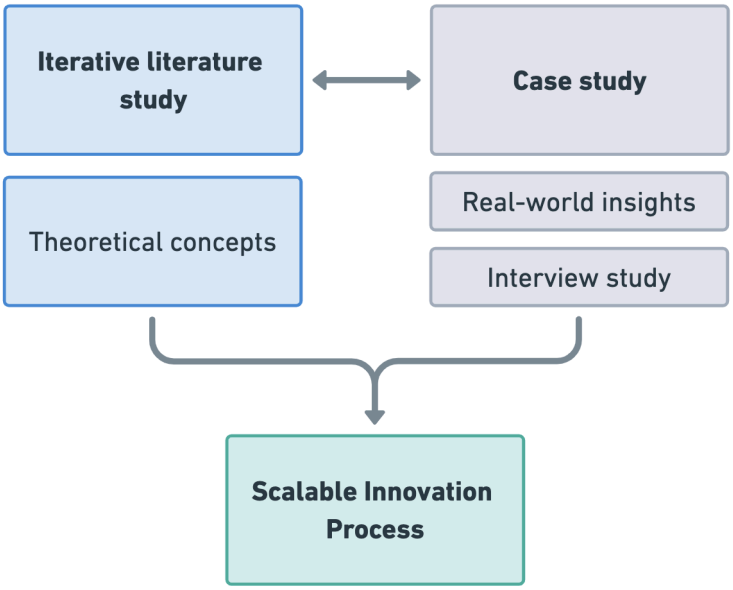


Figure 5. Conceptual research design (Own illustration).

3.2 Research process

The beginning of the research process included identification of the research area and determination of limitations. To ensure that common goals were set from both KTH Royal Institute of Technology and the case company, conversations and discussions were held with both organizations. To meet the interest of both the case company and academic stakeholder, the expertise from both sides were included in the process of facilitating the adoption of research questions.

In order to gain an understanding of already existing research, an iterative literature review was conducted. The interview study was conducted to build a perspective of the company's innovation process work. The semi-structured interviews were conducted to collect qualitative data on the company's way of working with new ideas and challenges around innovation process work. This included investigating whether innovation processes are utilized in practice, and if so, examining how they are implemented. Additionally, data collection from internal resources of the case study company was conducted. The documentation included annual reports, sustainability reports and internal presentations of projects that had been pitched and declined or implemented within the case company.

3.3 Literature review

A literature review was conducted throughout the study with the aim to discover new perspectives on the subject and to find strategies to develop a scalable innovation process to use in the grocery retail sector. To develop a comprehensive understanding of the research area, the literature review combines different perspectives, adopting a broad scope to create a robust foundation of the definitions and key concepts of innovation process work. Drawing from various fields, including innovation management, sustainability, historical innovation processes and retail innovation, the review aims to provide a thorough framework upon which innovation processes can be evaluated and refined.

The insights acrossfields come from several first-hand sources, including academic papers, practical industry reports and real-world literature examples of scalable innovations in the retail industry. By

exploring existing academic literature together with practical case studies, the aim was to identify knowledge gaps, challenges and opportunities within the sector.

3.4 Case company

This study was conducted in collaboration with one of the largest grocery retail companies in Sweden with over 5000 employees. The company has both company-owned and franchise-owned brands operating physical and digital stores and storefronts. They have an in-house wholesale company and produce private label products, as well as a wide range of external brand FMCG. The case company has national coverage in Sweden and has been operational for more than 50 years. The high-level organizational structure is visualized in Table 1.

Table 1. High-level organizational structure of the case company.

Business areas	**Description**
Corporate group level	The headquarter are in charge of strategy, HR, IT, finance, legal work, business development etc. Centralized management work with high-level development for the whole group
Grocery retailer (the case company)	Management work with specific development efforts for the group-owned and franchise-owned brands, aligning with the corporate group level directions
Group-owned grocery retailers	The brand of physical and digital stores owned by the group.
Franchise-owned grocery retailers	The brand of physical and digital stores owned by franchisers
Wholesale operations business	Logistics, procurement, development and production of private label products

3.5 Data collection

In order to gain knowledge on how professionals in the grocery retail industry perceives innovation processes, semi-structured interviews were conducted (Table 2). According to a methodological reflection by Ruslin et al. (2022), this interview format is beneficial in qualitative data collection as it allows for flexibility and adaptability whilst still driving the interview in the direction needed to focus on the research questions.

The aim of the interview study was to build a perspective on the typical innovation process work within the company, and to identify key challenges. Interviews were conducted with employees in the case company across different business units e.g., to provide a broad overview of the company's overall strategies, processes and challenges around innovation. The semi-structured interviews were centered around how innovation processes are utilized in practice in different departments, and which challenges occur when working with innovation today. Here, several employees within product development and retail supply chain were interviewed, focusing on understanding how they work with innovation and what clear processes exist, if any. This made it possible to capture both the strategic aspects of innovation processes, and how these are put into practice and implemented at different levels within the company group.

Table 2. List of interview objects.

Number (#)	Job title	Industry	Duration (min)
1	Sustainability Manager	Grocery retail	42
2	Business Development Manager	Grocery retail	21+30
3	Responsible for Purchasing & Logistics	Retail supply chain	54
4	Product Developer	Retail supply chain	22
5	Purchasing & Logistics Employee	Retail supply chain	36
6	Business Developer	Grocery retail	48
7	Packaging Developer	Grocery retail	31

In addition to the qualitative interview data collection, confidential internal documentation from the case company has been collected and thematically analyzed quantitatively to complement and nuance the interview study and broaden the analysis.

3.6 Data analysis

Initially the qualitative data from the interview study was transcribed by leveraging software, as well as manually, to decrease the risk of errors and knowledge losses in the data analysis process (Nowell et al., 2017). Thereafter, a thematic analysis (Table 3) of the material inspired by a six-phase method by Lincoln and Guba (1985) was conducted to compare and contrast the material before clustering. By taking several steps including familiarization with the data, identification of trends detected across the interview material and clustering into themes, a rigorous analysis was conducted.

Table 3. Thematic analysis method

Analysis phase	**Task**
Processing of raw data	Transcriptions conducted by software and manually
Data familiarization	Cross-referencing of the collected data, summaries of key information identified
Identification of trends	Trends spanning challenges, innovation process routines, ad-hoc tendencies etc were detected and analyzed across the material to ensure consistency and reduce interpretation bias
Clustering into defined themes	Refining of trends into 8 high level themes and cross-checking with raw data

Creswell (2017) suggests mixed method analysis, where quantitative data is integrated and weighted together with qualitative clusterings, allows for more nuanced analyses. In this study, confidential internal documentation from implemented innovations, long-term strategies, sustainability reports and key performance indicators for determining project success, revenue streams and specific yearly goals have been cross-analyzed with the findings from the interview study. As the interview objects work in different business units, differences in challenges addressed could be compared to further gain understanding in innovation process practices across the company to identify process variations.

4. Results: Innovation challenges in practice

In this chapter, the eight core themes identified in the data analysis process are presented, highlighting the common structures and challenges which characterize the reactive and ad-hoc innovation process work in the case company (Table 4).

Table 4. Themes and categories identified

Theme	**Category**
Internal and external collaboration	- Cross-functional teamwork - Alignment across teams - Partnership with external suppliers - Collaboration tools - Synchronizing project work/meetings
The role of project leadership	- Decision-making - Project oversight - Resource management
Case-driven innovation processes	- Business value proposition - Feasibility assessment - Limitations of radical innovations
Pilot projects	- Test and validate business value
Ad-hoc innovation and external pressure	- Adapting to market trends - External pressures - Reactive approaches
Company culture	- Challenges in large organizations - Perception of innovation
Customer behavior	- Slow customer adoption - Understanding customer needs - Customer feedback
Revisiting unprioritized ideas	- Idea repositories

4.1 Internal and external collaboration

A recurring theme from the interviews revolved around collaboration, both internally between departments, and externally with suppliers and other stakeholders.

According to the interviewees, unstructured internal collaboration within the organization presents challenges given the lack of standardized means of communication, which limit the organized innovation work within the company. The challenges of keeping a cohesive teamwork often leads to siloed efforts, where daily operational work is prioritized, reducing the focus on driving innovation work. Additionally, the interviewees highlighted how challenges commonly arise in projects that involve cross-departmental collaboration where each team contributes to different parts of the process. The collaboration does not always happen seamlessly since departments have different routines, approaches and procedures. Hence, teams in the organization work in silo, rather than through an integrated and synchronized method, resulting in miscommunication when new ideas and solutions are both suggested and developed.

To avoid miscommunication and to help track the processes of various projects across teams, one interviewee spoke about a newly implemented management tool. This tool provides the employees with a project overview of all the new launches and supplier changes, creating opportunities to track the product throughout the process, ensuring that all stakeholders fulfill their responsibilities. However, the interviewee emphasized that the tool’s success relies on its full implementation, proper utilization across departments, and engagement and responsibility from individual team members. Some teams are consistent in using the tool, while others don't prioritize it, resulting in challenges in communication and coordination.

Another interviewee mentioned that their department has started weekly reconciliation meetings to synchronize the team's project work. At first, the interviewee saw these recurring biweekly meetings as redundant and unnecessary, but once the team sat down they realized the value for the identification of needs and challenges within the group, strengthening the collaborative spirit by collectively problem-solving. This insight iterated the value of recurring team

meetings to fill an important alignment function, even if the value was not initially recognized. One interviewee stated:

“Just by seeing each other in these meetings you gain insight into much more than if you worked in isolation”

Even though internal collaboration is not always free from challenges, it is clear that the interviewees emphasize the importance and advantages of having a well-functioning collaboration between departments and teams. One interviewee expressed:

"I think it's important to work together in cross-functional teams, otherwise you risk getting stuck in your own part and becomes isolated from the whole”

Insights from several interviewees emphasized how ideas and thoughts often develop naturally during meetings and synchronizations with colleagues. During these meetings, the discussions may begin with a specific topic and evolve to new insights and ideas as other topics get explored, leading to innovative solutions and deeper understandings.

In addition to internal collaboration, the interviewees addressed the importance of collaboration and communication with external stakeholders. The case company works closely with suppliers to innovate and adapt products to meet evolving needs and trends. These supplier partnerships are essential for ensuring the flexibility and product range required for the company's sales. Therefore, the suppliers play a crucial role, not only by providing the products but also in driving innovation to align with market trends and customer demands.

Additionally, the interviewees highlighted the importance of maintaining close relationships with other external stakeholders, e.g., packaging producers and recycling industries. These stakeholders have valuable insights on sourcing new materials that meet both sustainability goals and operational requirements. However, the company sometimes struggles to align external innovations from suppliers into its internal processes. This can slow down the implementation of supplier-driven innovations and potentially create friction between the two parties.

4.2 The role of project leadership

The interviewees mentioned how effective project leadership is important for driving innovation. Specifically, it is important to have project leadership with common approaches of frameworks for guiding teams. While efforts are being made to align functions, the lack of this can result in important details falling through the cracks. This can also result in unclear expectations and misaligned objectives, causing teams to work in silo rather than collaboratively.

To ensure innovation initiatives and projects are aligned with the group company's strategic goals, interviewees emphasized on the importance of having a strong leadership function who are well aligned with the group company's overall strategy. One interviewee commented:

"We need to have a cohesive leadership to ensure that our innovations are not carried out in isolation, but rather connected to our overall company goals"

According to interviewees, initiatives that are dependent on digital services or IT, require collaboration with the corporate group level. For significant projects, such as launching a new retail concept a project manager from the case company's project team has to collaborate with for example a project manager from IT at the corporate group level. Then IT handles the technical components, such as the development of digital infrastructure, while business development from the case company handles the operational aspects, including rollout to stores, coordination of resources and communication with staff.

However, when several teams own different workstreams it is important to avoid misalignment and fragmented efforts. Ensuring leadership roles collaborates and commonly oversees the project, aligns to the project goals and streamline processes is therefore essential. One interviewee highlighted the challenge:

"It is important with a centralized leadership that can ensure all departments are aligned when driving innovation projects"

The case company's project management plays a crucial role in driving big strategic projects which do not require allocation of IT resources from the corporate group level. These initiatives are sometimes extensive and strategically important, usually requiring decision making from the corporate group level to get started. After approval, the case company's project management determines the key focus areas for the upcoming year and sets the annual strategic goals for the projects.

It is then up to the case company's project teams to ensure the work moves on according to plan. Although most major initiatives are predetermined and planned, unexpected needs arise that fall outside the scope of the strategic plan. Therefore, leadership guidance is important in these scenarios, ensuring pre-planned and unforeseen projects are aligned with the company goals.

4.3 Case-driven innovation processes

A recurring theme from the interviews is the frequency of case-driven innovation work at the case company. Typically, decisions and initiatives are commonly determined by feasibility and generated business value. A manager explained that roughly eight out of ten ideas which are brought up are approved and implemented, indicating an incremental-first driven approach to innovations. Innovations are solely investigated and piloted if they are likely to have a high success rate and present a clear business value. Success rate is the proportion of ideas or projects that can be successfully implemented. These incremental innovations are characterized by having lower risk and being easier to implement, rather than radical innovations which have higher risks of failure.

However, one interviewee bringed up that this approach limits the exploration of more radical innovations since the focus primarily lies on meeting the immediate operational needs, and avoiding the risk of the project not reaching its full completion. The interviewee clarified:

"There is not always enough time or resources to invest in projects that cannot immediately demonstrate a clear business value, resulting in more risk-taking ideas not receiving the support they need to be implemented"

This, according to the interviewee, means that innovations requiring significant investment often get deprioritized as they cannot prove business value in a foreseeable future.

Furthermore, it is evident from the interviews that new ideas are evaluated from a risk and return perspective. It is explained that, if an idea requires a large investment, it is required to provide significant business value to be approved to be further iterated and developed. However, if there is no clear evidence that the investment will pay off, it is difficult to get the resource allocation required to continue the project.

4.4 Pilot projects

When launching something new, the interviewees explained that pilot studies are most often conducted before rolling it out into the entire chain. The purpose of the pilot is to test and validate the hypotheses regarding the business value, and the expected benefits of the initiative. If the pilot shows that the results do not meet the expectations, it gets canceled, and the focus shifts to other initiatives. The pilot assessment thus acts as a gate to confirm the sufficient grounds to invest in the idea and its potential to create value. If the business value cannot be proven during the pilot, it may lead to adjustments in the solution, e.g., extending the pilot study, or in some cases, ending the project altogether. One interviewee stated:

“In this way, we ensure that only the initiatives we really believe in and see value in are rolled out widely within the chain”

For some initiatives, where the direct business benefit is not entirely obvious during the pilot phase, it is hard to say with certainty that it will pay off within a foreseeable future. Some initiatives require a long-term perspective to assess the business value, where the potential increases over time, rather than through immediate results. Other initiatives which have been partly developed previously and had budget allocation have a lower threshold for proceeding as part of the associated costs have already been covered, and it would be an alternative cost to ramp it down without properly assessing the long-term value potential.

4.5 Ad-hoc innovation and external pressure

In an industry characterized by a strong reliance on customer needs and market trends, some of the case company's initiatives stem from external pressures and needs that emerge suddenly, rather than anticipated improvements accounted for in the long-term business strategy.

The interviewees provided insights on how the company's innovation process is driven by external demands. Many of the innovations related to physical products are iterations of strategic moves to comply with regulatory changes and adjust to supplier demands. For instance, biodegradable packaging has been adopted in response to sustainability regulations regarding plastic usage, based on how well they comply. Hence, this is an example of a reactive innovation, that according to the interviewee ensures the compliance needs driven by external factors. Reactive initiatives are often seen around the organization and in these cases, the interviewees argued that it is important to anticipate the long-term compliance consequences and alternative costs of these strategic decisions.

According to a business developer at the case company, most innovation initiatives are predetermined and planned, but unexpected needs occasionally arise that were not included in the plan from the start. These ad hoc initiatives are evaluated on a case-by-case basis, considering the available resources and the business value of the project. This determines if the initiative can be proceeded with right at the time, or if it should be added in the process for the upcoming year.

4.6 Company culture

Results from the interviews show that representatives of the case company rarely perceive themselves as innovators. One interviewee described how they do not see their business as being directly focused or centered around innovation. Rather, the greater focus is on developing long term strategies addressing organization changes over time, with initiatives which they rarely think of as a form of innovation. Instead, the business is focused on continuously providing products and services for customers in a standardized way as primary customer offering. Additionally, interviewees stated that a

strong caution towards taking risks is somewhat embedded in the company's culture. However, this can sometimes leave less room for innovation.

The interviewees also mentioned the clear organizational leadership structure, resulting in that decision-making often comes from the higher levels of the organization. Hence, this organizational chart makes it difficult to bring initiatives forward for less senior employees in the organization. Additionally, it risks contributing to a slower realization of ideas as these need to be approved by several different levels of leadership before they can be implemented. One interviewee clarified:

"Being a large organization with established processes and certain steps can feel slow and bureaucratic, making it challenging to be as agile as innovation work sometimes requires"

For larger projects, the processes can be accelerated through internal prioritization when there is a strong business need. Otherwise, the projects are required to follow standardized procedures that often take time due to the many dependencies and stakeholders involved in the decision-making process.

4.7 Customer behavior

Another aspect of the difficulties with innovation within the grocery retail industry is the pace of change, which is generally slow compared to other retail industries. According to several interviewees, this is due to the fact that customer behaviors are deeply rooted and take a long time to change in grocery retail. An example of this is when a competitor started testing the technology of mobile scanning a couple of years back. Although the technical solution worked, there was no receptivity or interest from the customers. As a result, the innovation did not succeed at that time. This illustrates how a technically feasible project does not always mean success if the customer behavior does not change at the same rate. One interviewee expressed:

"We can drive customer behavior to some extent, but at some point, it also has to align with the customer's level of maturity"

Interviewees expressed that innovation work in grocery retail requires high responsiveness to customer needs and adherence to gradual behavioral changes in target segments. The interviewees also argued that companies operating in the grocery trade must not only consider the technical and business feasibility of their innovations, but also the cultural aspect of customers, which sometimes slows down the implementation of new, more radical ideas.

Additionally, the slow pace of customer adoption makes it challenging to be a pioneer and launch new solutions. In some instances, it is rather more successful to wait until the market is ready. This also influences the company's strategy to ensure that innovations have clear value creation for the company in the foreseeable future.

A large customer focus is crucial to ensure that innovations meet customer needs. Yet, interviewees argued that the customer dialogue requires improvements to further understand the target segments. The customer feedback gives the company a general view of customer perceptions and helps identify areas of improvement. One interviewee argued:

“It is important to have a much closer dialogue with our customers all the time to ensure that we are doing the right things”

Currently, the dialogue with customers is mostly collected in an ad-hoc and informal manner through customer feedback in the grocery stores, as the store personnel and franchisers often have a closer direct contact with the customers. However, the process of collecting and leveraging feedback is unstructured and dependent on who is running a specific project. Sometimes the customer dialogue is included in the planning for specific initiatives, but there is a lack of overall structure to continuously integrate customer feedback into both strategic efforts and innovation work.

4.8 Revisiting unprioritized ideas

According to the interviewees, there is no clear structured process for logging and following up on the ideas that are not immediately prioritized in the case company. Projects are often assessed by business value, and put aside if they are not chosen to proceed with. Moreover, the chances of the projects being revisited and lifted again are low. Some larger ideas which are not prioritized are stored in a backlog tracker by certain teams, whilst smaller initiatives and ideas are not stored or followed up.

5. Results: Proposed innovation process

In this study, a scalable innovation process in six phases tailored to the case company's challenges and needs has been developed (Figure 6). By identifying challenges in the grocery retail innovation process work from literature, the case study and interviews, the process has been designed to allow for both smaller ad-hoc innovations, larger projects and storing ideas in a structured way.

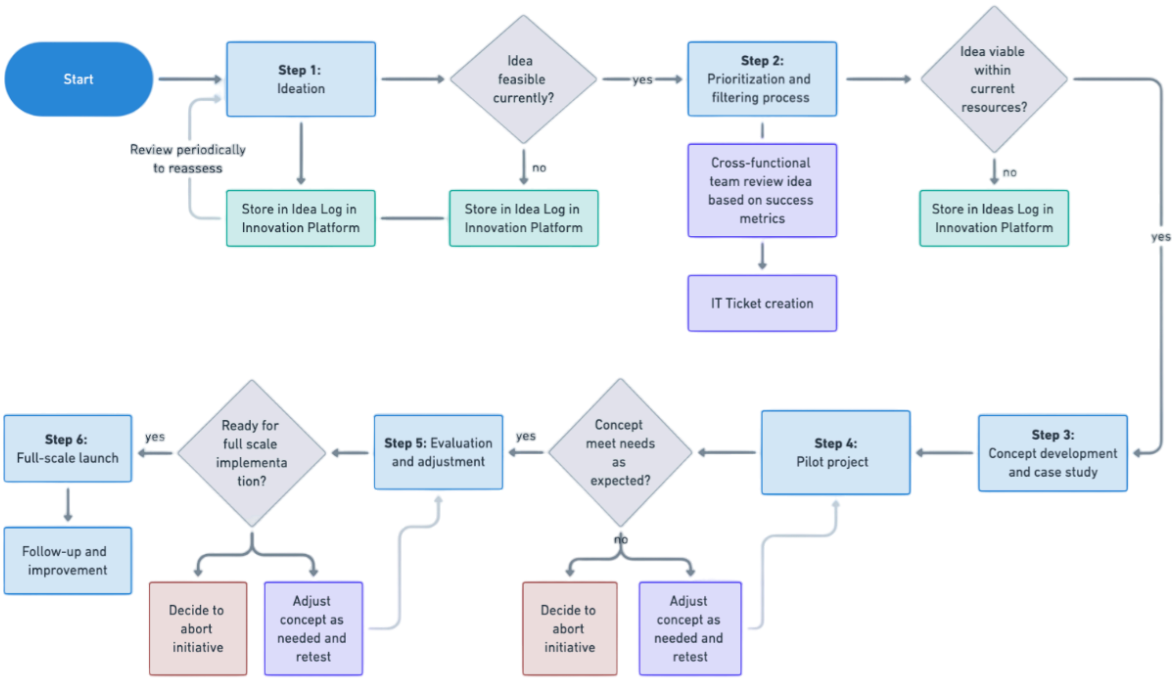


Figure 6. High-level overview of the Scalable innovation process.

The six phases of the scalable innovation process are as follows:

1. Ideation

   The purpose of the ideation stage is to log ideas which occur all across the organization, stemming from both internal and external stakeholders. These can be in the form of feedback from employees or customers, improvements of work situations or something else. The ideas and data points on ideas such as product development, packaging and customer experience innovation are registered and categorized in a digital idea

storage solution in the Innovation Platform. This mitigates the risk of losing ideas in an ad-hoc environment.

2. Prioritization
   A panel with representatives from different departments and brands within the organization will discuss these ideas against doability measurements (e.g. strategic relevance, resources required, team bandwidth and potential customer value). The most lucrative and doable ideas get advanced in the funnel, and the rest are stored as later time possibilities in the Innovation Platform. This stage ensures that the innovation work aligns with the overall strategy.

3. Development of concept and case study
   Ideas are further iterated through concept development where they are further realized, and analyses are made to address challenges, needs and constraints. Conceptual prototypes can be developed. This allows a more in-depth investigation of the viability and scalability of the innovation.

4. Pilot project
   In this phase, test groups such as internal teams, customers or franchisers try the concept in a pilot and provide feedback. This process enables testing the concept before complete rollout which reduces the risk of failure and ensures that the innovation addresses the needs of the target audience.

5. Evaluation and adjustment
   Results and feedback are reviewed and tweaks can be made according to it. This phase is an iterative process to improve the concept, preparing for full-scale implementation.

6. Full-scale implementation
   The final stage of the innovation process, if approved, is full-scale rollout which takes place across the company or in selected stores. This step includes training and communication with stakeholders to ensure a smooth implementation, making the innovation sticky.

6. Discussion

This section discusses the results and the developed scalable innovation process in the light of the research questions of the study:

- *RQ1. Which key challenges and features characterize the innovation process work in a grocery retail company?*
- *RQ2. What does an innovation process look like to be scalable and applicable across different business units in a grocery retail company?*

Additionally, the results are discussed and compared to literature about innovation work and established innovation processes. The aim is to reflect on how the study makes a contribution to the general comprehension of the work of innovation by utilizing structured processes in the grocery retail industry, as well as addressing opportunities for improvements. The discussion considers the specific challenges that exist in grocery retail, an industry distinguished by low margins, slow changes in customer's behavior and high competition.

6.1 Ad-hoc innovation work and organizational silos

Research describes innovation work in the retail industry as ad-hoc and case-driven, with a focus on early idea-generating initiatives such as brainstorming, and overall delivery of short-term solutions inspired by competitors rather than structured, scalable processes based on long-term revenue potential internally in line with the company's strategic plan (Sheu & Lee, 2010). The interview study supports this picture as innovation work in the case company is driven by needs and challenges in smaller teams across the organization. These efforts thereafter result in local initiatives in different business units across departments, rather than through a standardized innovation process for the whole group. As a result, this contributes to fragmentation in the work on innovation as it depends on the initiating teams' own ability and capacity to drive and implement innovations to a large extent. Another challenge caused by organizational silos is the fostering of a poor innovation culture internally. Olsson (2019) argues that employees feel unseen and fail

to realize their potential idea generators, due to the silo effects of their narrow contact net internally in the daily work.

Furthermore, literature on innovation work in retail differs from the case company's practical work in the aspect that the case company has an internal ambition to keep ideas in a backlog for future implementation, despite the documentation process not being systematic today. This crucial step in innovation work is not highlighted in literature around retail and innovation processes. This indicates that there is a desire to eliminate certain ad-hoc processes described in the literature, but that practical obstacles, such as lack of resources and organizational silos, still greatly affect the work and create barriers for structured innovation process work.

To address the challenges, the developed innovation process for the case company includes a routine for storing ideas and concepts in an Innovation Platform in the digital infrastructure, in line with Riege (2005) who argues that a robust digital infrastructure likely reduces knowledge sharing barriers. Furthermore, the process involves a manual screening of new innovation ideas by a panel of representatives from different departments and brands. These steps aim to ensure that the organizational silo tendencies related to innovation work decrease, and that there is a more transparent and collaborative view on innovations across the company's business units. Additionally, a structured process for reviewing ideas and assigning ownership to develop innovations throughout the process ensures less redundancy and double work.

6.2 Stakeholder collaboration

In retail, working with external stakeholders like suppliers and customers is crucial to strengthen the innovation capabilities and ensure long-term competitiveness for companies. The literature shows several examples on external stakeholder relationship building to strengthen innovation capabilities, e.g, the successful work with co-innovation labs by Mercadona (Albors-Garrigos, 2020). Moreover, Pantano and Giuseppe (2012) describes how customer-centric solutions and methods, like loyalty programs and structured collection of customer feedback, drive the development of more customer-adapted solutions. Additionally, Barile et al. (2018) highlight how IoT and digital systems can strengthen the

collaboration with logistics partners via real-time monitoring and communication to structure the flow of information.

Similarly, the interview findings confirm that the collaboration with suppliers is important ensuring products meet changing market needs, adaptability and flexibility. This in turn, is one important driver for innovation according to the interview study. While suppliers are essential contributors, ideas from other departments in the organization, stakeholders, other external stakeholders and feedback from customers, are also valuable.

Taking these into account to the innovation process will introduce fresh perspectives and address challenges from different angles, that will ensure the process is comprehensive and customer focused. Therefore, throughout the entire developed innovation process, it is important to be open for ideas coming from different sources. Through the innovation platform, ideas can be stored regardless of the source. Ideas that are relevant immediately can be reviewed, but if they are not relevant at the time, they get saved for future consideration. Results from the interview study showed that employers rarely perceive themselves as innovators. This openness to idea gathering, could therefore foster a collaborative culture at the company, making it easier to contribute to innovation, and in the end get the feeling of perceiving themselves as innovators.

6.3 Incremental vs radical innovation focus

According to Evans (2011), retail is characterized by incremental rather than radical innovation work. As a result, there is a risk to miss strategic opportunities and therefore face great opportunity costs over time, as well as lost economies of scale effects. This pattern is also reflected in the case study, where innovations must show potential for short-term returns in order to be prioritized internally within the company. The limiting factors of innovations including short team bandwidths, lack of structured processes and budget constraints make it difficult to allocate resources to more experimental projects with uncertain outcomes, and therefore realizing certain more radical innovation initiatives. Furthermore, the interview study highlights that ideas are more likely to be prioritized if they have a clear business case in a reasonable timeline, causing radical ideas to rarely gain traction. At the same time, the interview study showed that there is some flexibility internally for testing

smaller initiatives, but these are often missed due to a lack of storing and realizing the ideas in a structured process.

To ensure that both incremental and radical innovation ideas are assessed against the doability measurements internally, the developed innovation process allows for storage of all ideas in the Innovation Platform, and categorization based on type of idea. Furthermore, these ideas will be screened by the panel for potential to be realized, ensuring a more structured assessment. As the case company has a clear strategic focus on incremental innovation first, rather than being an early adopter, the process is hence tailored to allow for incremental ideas to continuously be the primary focus area, whilst still allowing for radical innovations to be stored and later developed through the Innovation platform.

6.4 Adhering to customer needs

In the grocery retail industry, customer satisfaction is directly correlated to brand loyalty and how frequent customers are in interacting with the brand. Hence, the grocery retail industry is customer-centric by nature. Understanding customer needs and translating these into requirements and then feasible solutions and products is highly important for the companies to remain competitive (Rodriguez et al., 2016). However, the study suggests that challenges in obtaining and implementing customer feedback in the innovation work in a continuous way to balance the larger trends and market shifts and the immediate customer needs and wants. To ensure that this smartness is captured and implemented in a structured way, the developed innovation process integrates the customer demands throughout the full process in an interactive and dynamic way. Moreover as Rivera (2021) discusses, IoT and RFID tools may be leveraged to understand customer interactions with innovations in pilots and case studies conducted.

6.5 Innovation processes comparison

The developed scalable innovation process is based on research on established innovation processes utilized across several industries, and then tailored to the case company, meeting its customer- and market-specific needs and challenges. By leveraging strengths from

different methodologies, the developed process is flexible and robust enough to be scalable across business units to be applicable for developing and implementing different types of innovations whilst addressing key challenges of silo work and lack of structure and documentation in the innovation work of today in the company.

The scalable innovation process developed aims to be flexible and therefore adaptable to specific needs depending on the type of innovation and business unit within the case company. Moreover, it is structured to drive the projects in a consistent direction to ensure balance between modularity and structure. The process is collaborative and encourages transparency through an accessible Innovation Platform as well as a cross-functional panel review phase to break the silo tendencies. The collaborative barriers are highlighted by Olsson et al. (2019) as well as Riege (2005), agreeing that reduced collaboration has a negative impact and causes fragmentation in innovation process work. By leveraging a scalable innovation process, the case company can find a balance between prioritizing innovations for short-terms operational wins and long-term strategic success, adhering to customer needs and reducing alternative costs of missed opportunities due to reactive innovation behaviors.

By having phases and gates, the process is influenced by the linear Stage Gate process. Stage gate is optimized for product development and R&D, offering a highly structured methodology with clear gates before progressing to the next stage which allows for clear documentation and ownership throughout the process. In similarity, the developed innovation process is centered around risk mitigation through clear decision-making gates. However, to ensure that the process is flexible and scalable enough to meet the needs of the grocery retail case company, the process has a higher level of modularity.

Moreover, by including testing and concept adjustment, the process is similar to Agile Innovation where rapid feedback and continuous improvement are prioritized. Therefore, innovations are scaled after smaller scale validation through case studies and pilots are completed, which increases the adaptability of the process and reduces the risk of allocating too many resources before seeing a clear business value.

Similarly to Design Thinking, the developed innovation process has a strong focus on cross-functional collaboration to meet customer needs in the final full-scale implementation phase. As a panel screen and assess the ideas, insights from multiple perspectives are lifted.

In summary, this study demonstrates the importance of utilizing a structured innovation process in the grocery retail industry to lay a solid foundation for proactive innovation work. Utilizing the process will likely enable the case company to enhance their innovation capabilities and create short-term and long-term value across business units.

7. Conclusions

The study focus was to identify the key challenges for structured work following an innovation process in a case company operating in grocery retail, and developing a scalable process to meet the needs across business units.

As highlighted in the discussion, key challenges for innovation process work in the grocery retail organization includes lack of structured means and methods of communication and collaboration between internal teams, as well as with external stakeholders. This causes the collaborative efforts to be inconsistent. Additionally, without clear management ownership for capturing and implementing innovations, there are risks for dead stops as the projects are not advanced. Case-driven and ad-hoc innovation work result in silo structures where certain teams build knowledge on how to develop and implement ideas specifically relevant to their sole team challenges. This may cause the long-term lucrative ideas to fall in status compared to the short-term doable solutions. Additionally, the element of customer behavior adaptability plays a big role in the willingness to adapt new solutions stemming from innovations in the environment facing customers.

To develop a scalable innovation process which can be utilized across the organization, there are key focus areas ensuring that the process is flexible and modular enough to allow for use across business units, realizing the specific needs and types of ideas in different parts of the organization. Additionally, the innovation process needs to ensure that innovation ideas are validated and developed into feasible concepts before being implemented to de-risk and ensure that the right solutions are being developed at the right time to align with the business strategy. Moreover, to ensure that the process is feasible for different stakeholders across the organization, the usage of cross functional panels to assess ideas allows for natural integration of different perspectives and less silo work in teams and departments. Furthermore, by storing the ideas in an Innovation Platform which can be accessed across the organization, there is an increased transparency and also allows for inspiration across teams.

This study contributes to the research field of innovation work in retail environments by providing a structured overview on key challenges for working with innovation in a scalable way in the industry, addressing both operational requirements and strategic implications. Moreover, it addresses how innovation processes can be developed to be scalable across business units and for ideas spanning from smaller initiatives to large implementations stemming from innovations. Lastly, the study addresses the importance of merging insights from theory and practical work for empirical validation of theoretical concepts in practical operational environments.

8. Limitations and future work

8.1 Limitations

The study researches one grocery retail company in Sweden, and therefore the results and recommended innovation process is tailored to fit the identified needs of this sole actor. Hence, the transferability to other retail companies in both Sweden and other markets may be limited given the differences in organizational structure, work culture and challenges for innovation occurring. Furthermore, given that the interviews have been conducted with a limited number of employees, it is likely that there are many more perspectives and ways that the current innovation process work in the case company are perceived that have not been addressed in the study.

8.2 Future work

Future studies should identify success metrics and investigate the implementation of the suggested innovation process across these metrics over time. This could be done by qualitative measurements and quantitative tracking of innovation deliveries for projects which have been developed and implemented leveraging the developed innovation process. Furthermore, the results could be benchmarked against innovation processes in other retail companies in Sweden as well as in other markets and industries to build a broader perspective on required adjustments to tailor innovation processes to different companies. To further research the case company, research on how the suggested innovation processes connect to the overall strategy work for long-term success could be executed to identify synergies. To ensure a successful implementation of the innovation process, the digital environment for the Innovation Platform for idea logging should be developed by the case companies IT department.

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APPENDIX: Interview guide

The following interview guide was used as a base for the semi-structured interviews in the study. The guide was designed to provide the interviewees with the opportunity to elaborate on their answers, with questions adapted to each interviewees position at the company.

Introduktion

1. Är det okej att spela in?
2. Vilken roll har du på företaget? Titel? Avdelning?
3. Vad har du för relation till innovation på företaget?

Innovationsprocess

4. Har ni en strukturerad process för innovation?
5. Andra metoder/frameworks/manualer?
6. Kan du beskriva de typiska stegen från idé till implementering av en innovation?
7. Hur testas och itereras idéerna innan de implementeras i verksamheten?
    a. Vad är det huvudsakliga syftet med det?
8. Hur utvärderar ni framgång för innovationsinitiativ?
9. Vad händer med innovationerna som inte implementeras?
    a. Vilka beslut tas kring detta? Av vem/vilka?
    b. Finns det en logg?
    c. Plockas de upp vid senare tillfällen?

Utmaningar och yttre faktorer

10. Vilka utmaningar/hinder möter ni när ni implementerar innovationer
    a. Hur hanterar ni dessa hinder?
    b. Är hindren ofta interna eller externa?
    c. Har ni behövt ändra processer/metoder på grund av detta?
11. Hur påverkar regler och föreskrifter innovation?
    a. Andra externa påtryckningar/förändringar som påverkar?

Kundbehov

12. På vilket sätt påverkar kundbehov ert innovationsarbete?
13. Hur arbetar ni för att uppfylla kundens behov?
14. Hur integrerar ni kundbehoven i innovationsprocessen?

Innovationskultur

15. Vilken roll spelar innovation inom företaget och din avdelning?
16. Hur skulle du beskriva innovationskulturen inom företaget?
17. Skiljer sig relationen till innovation åt mellan avdelningar/nivåer i organisationsstrukturen?
18. Hur uppmuntras anställda att bidra med idéer?
    a. Utmaningar kring detta?
19. Hur skulle du säga att innovationsklimatet inom retail-sektorn ser ut?
    a. Är det lätt eller svårt att driva innovation?

Övrigt

20. Övrigt material som kan vara användbart i vår studie?
21. Något att tillägga?
22. Förslag på personer att intervjua?
23. Tack för oss!

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